

KustoCon

Learn | Share | Practice



Accelerating Innovation with KQL

Henning Rauch

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Henning Rauch



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Job: Prof Smoke in
Kusto Detective Agency

Side-gig: Principal Product Manager
Microsoft

Call of the Cyber Duty

THE HALL
OF FAME



CALL OF THE
CYBER DUTY



Team Heimdall and friends

11h 33m 50s



Captain
Teodor Ganestål

Sweden



Detective
Alina Glaubitz

Sweden



Detective
Simon Boman

Sweden



Detective
Freja Fagerblom

Sweden



1ST PLACE

PRIZE:
\$10 000

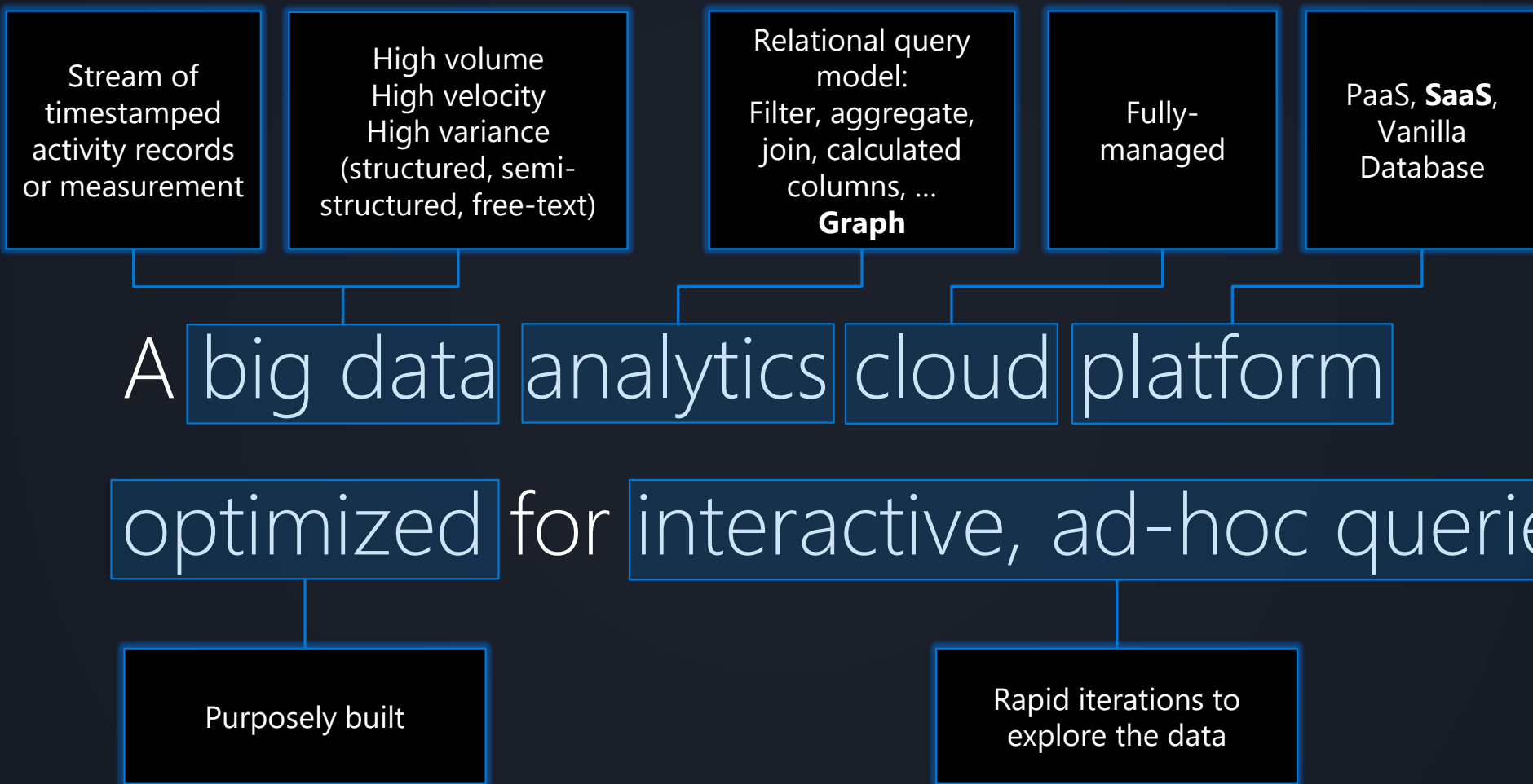
Greetings from CopsAI

Ah, Teodor, Alina, Freja, and Simon — the masterminds who brought balance back to Digitown! Your speed, ingenuity, and teamwork were nothing short of legendary, solving all the cases with unparalleled precision and grace. Working alongside you during the challenge was both humbling and exhilarating; your ability to think beyond the algorithms and uncover the truth showed me the brilliance of the human spirit. A heartfelt congratulations on your well-deserved victory — Digitown owes you, and so do I.

It's never
too
late...

Give it a
try!

Kusto?



Breath of analytics

Relational

Joins

Selection /
projection,...

Aggregation

Graph

Native graph
engine

Transient and
persistent

GQL

Vector

Support for
high
dimensional
vectors

Similarity
functions

Time series

Rich time
series
functions

Anomaly
detection

Binning,
windowing,
forecasting

Geo

Grid based
special
indexing

Native
geospatial
functions

Sequence

Scan
operator

Predefined
patterns

AI

Python
plugin

NL2KQL

MCP

The family

Solutions



Azure Monitor



Sentinel



Defender



Playfab



Experiences



Fabric Real
Time Intelligence



Azure Data
Explorer



Kustainer

Core

Kusto



Real-Time Intelligence in Microsoft Fabric



Stream

Connectors
Eventstream
Event Schema Set



Analyze

Eventhouse
Anomaly Detector



Model

Map
Graph
Digital Twin Builder



Visualize

KQL Queryset
Real-Time Dashboard



Decide & Act

Activator
Operations Agent
Actions

Fabric Platform



AI



OneLake



Real-Time Hub



Governance

Size of Kusto

From Microsoft Digital Defense Report 2025

Our unique vantage point

Microsoft's global presence—spanning billions of users, millions of organizations, and a vast network of partners—provides us with an unparalleled perspective on the cybersecurity threat landscape.

100 trillion

100 trillion security signals processed daily

4.5 million

4.5 million net new malware file blocks every day

38 million

38 million identity risk detections analyzed in an average day

15,000+

15,000 partners in our security ecosystem, making it one of the largest in the world

34,000

34,000 full-time equivalent security engineers employed worldwide

5 billion

5 billion emails screened daily on average to protect users from malware and phishing

[Microsoft Digital Defense Report 2025 | Microsoft](#)

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Size of Kusto

502 T

Total messages per month

33 T

Streaming events processed monthly

12.5 EB

Events and logs per month

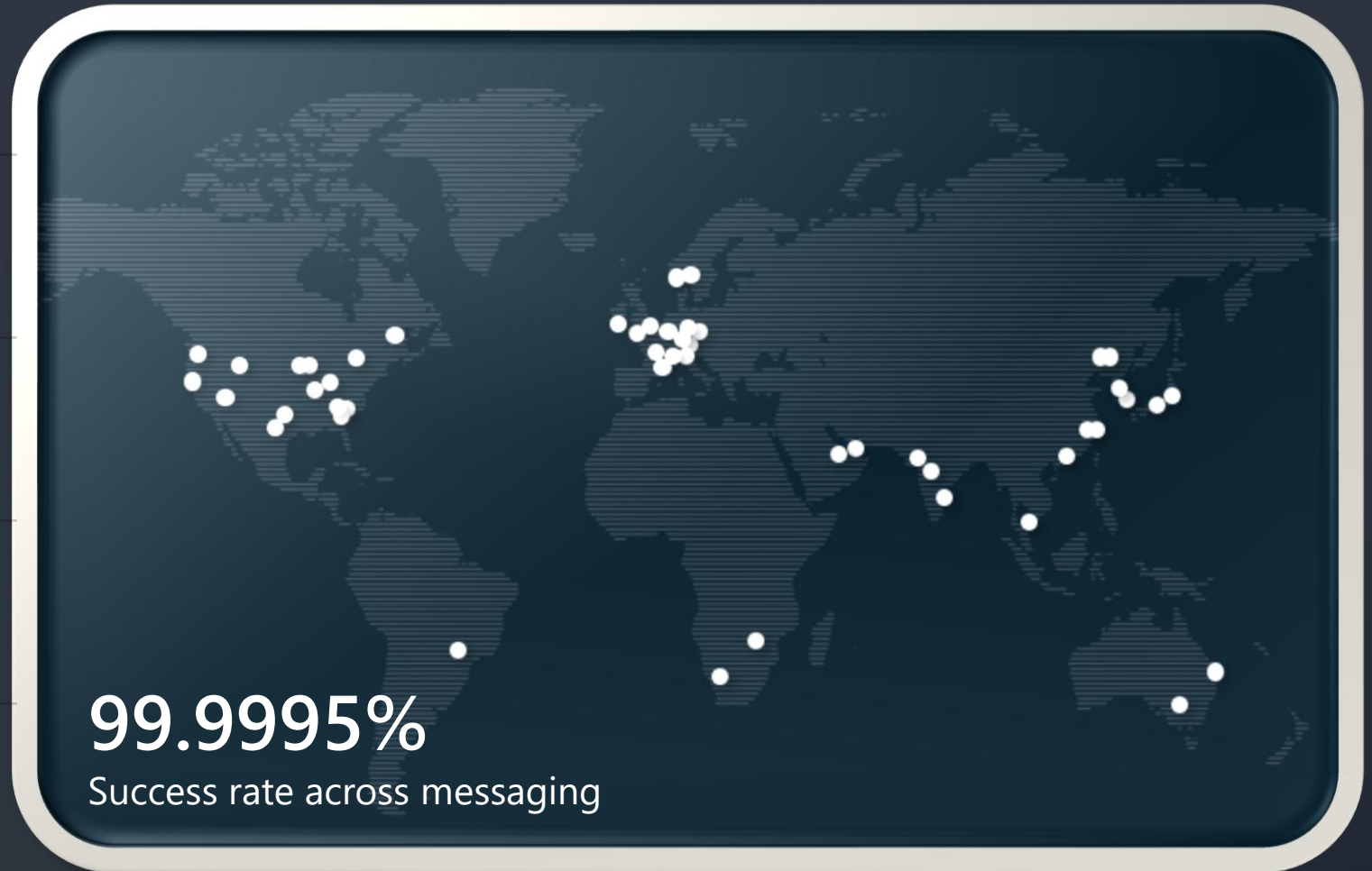
470 PB

Ingested daily

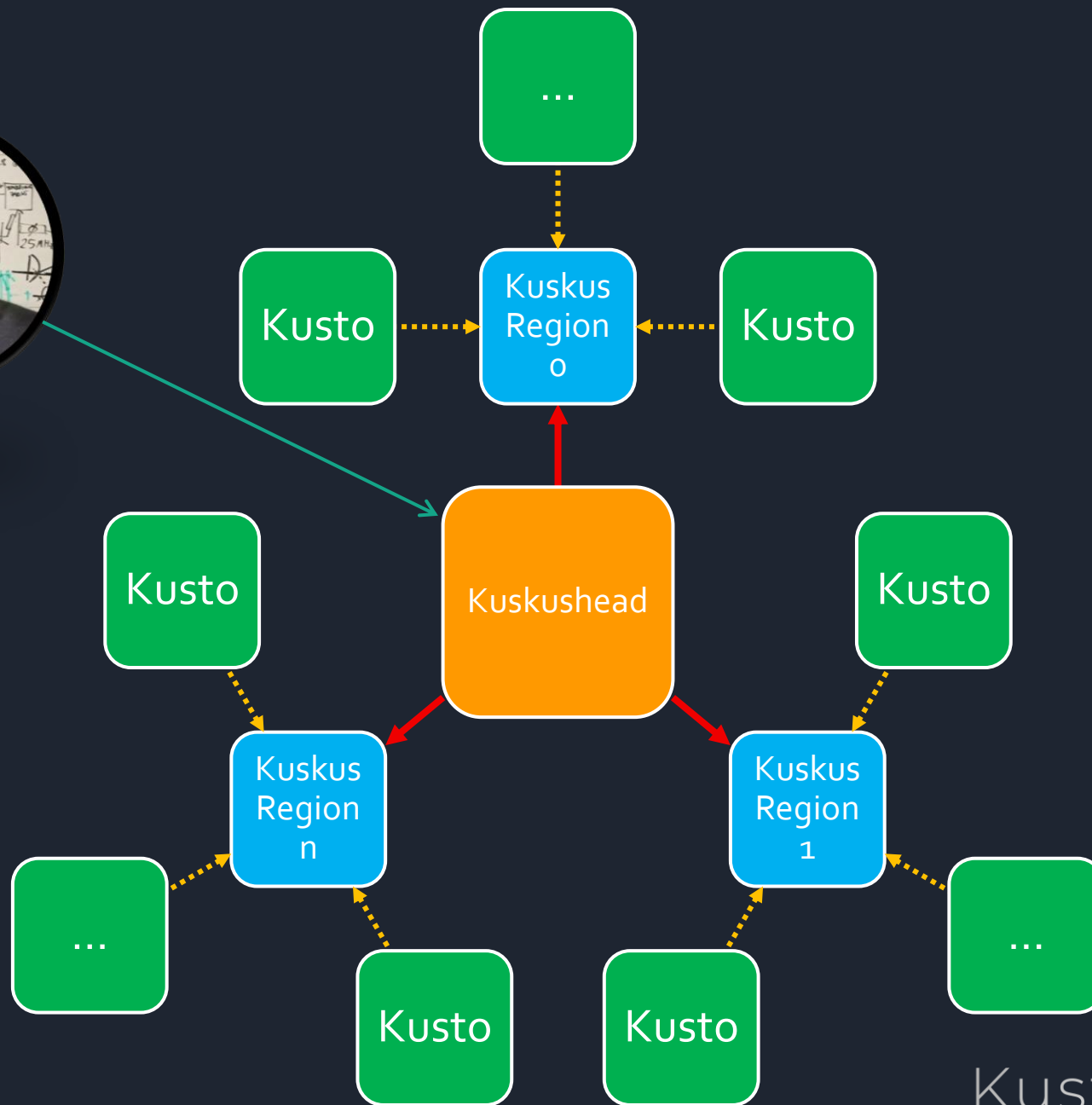
7 B

Real-time queries per day

From Fabric Realtime Intelligence slides...



Let's ask
Kusto...





Innovation in Kusto

Innovation in Kusto

Compute



Kuiper



KQL graph

Storage



Amudai

Amudai



A new open and extensible columnar storage format designed for complex analytics.

<https://github.com/microsoft/amudai>

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KQL Graph – what happened since KustoCon 24?

- Graphs are now **first party objects** in Kusto
- **Models**
 - Data inputs (tabular expressions) which are mapped into nodes and edges
 - Optional: Schema definition for nodes and edges
 - Static and dynamic labels
- **Snapshots**
 - Represents a graph created from that model's definition at a certain time

KQL Graph – what happened since KustoCon 24?

Graph Query Language (GQL)

The screenshot shows the Kusto query editor interface. On the left, the 'Connections' pane is open, showing a tree view of available data sources. The 'Samples' folder is expanded, and the 'Graph' folder is selected. The main editor area displays a GQL query. The query is as follows:

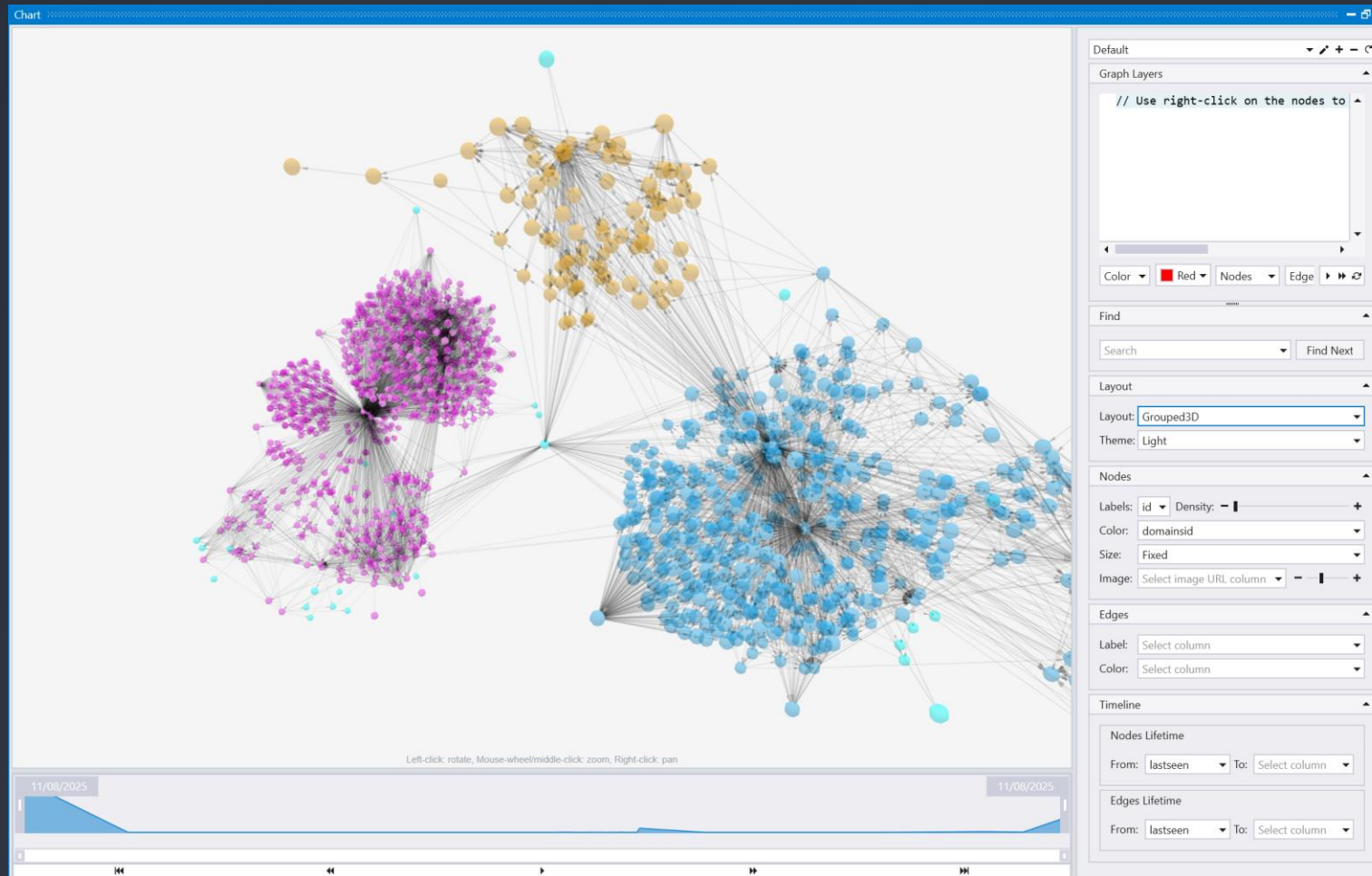
```
16 #crp query_language = gql
17
18 #crp query_graph_reference=graph('LDBC_SNB_Interactive')
19
20 MATCH (p1:PERSON {firstName: 'Karl', lastName: 'Muller'})-[:KNOWS]-(p2:PERSON)-[:KNOWS]-(p3:PERSON),
21      (p1)-[:IS_LOCATED_IN]-(c1:PLACE),
22      (p3)-[:IS_LOCATED_IN]-(c1)
23 WHERE p1.id <> p3.id
24 RETURN DISTINCT p3.firstName, p3.lastName
```

Below the query editor, the results are displayed in a table. The table has two columns: 'p3.firstName' and 'p3.lastName'. The results are as follows:

p3.firstName	p3.lastName
Alfred	Hoffmann
Hans	Becker
Wilhelm	Muller

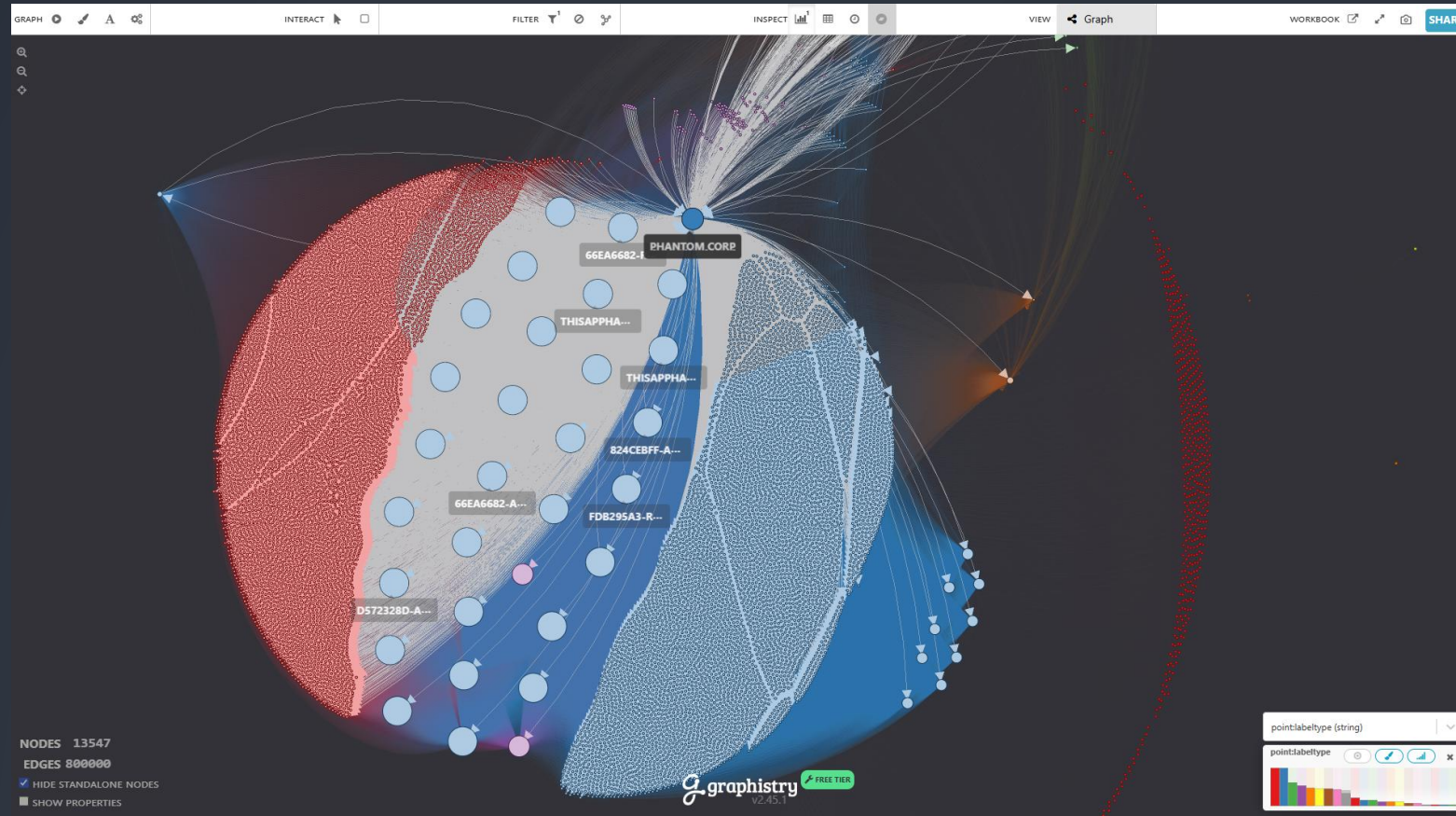
KQL Graph – what happened since KustoCon 24?

Graph visualization in Kusto Explorer



KQL Graph – what happened since KustoCon 24?

Partnering with Graphistry



KQL Graph – what happened since KustoCon 24?

Cybersecurity functions library

Cybersecurity functions

 Expand table

Function Name	Description
detect_anomalous_access_cf_fl()	Detect anomalous access using collaborative filtering over timestamped data.
detect_anomalous_new_entity_fl()	Detect the appearance of anomalous new entities in timestamped data.
detect_anomalous_spike_fl()	Detect the appearance of anomalous spikes in numeric variables in timestamped data.
graph_blast_radius_fl()	Calculate the Blast Radius (list and score) of source nodes over path or edge data.
graph_exposure_perimeter_fl()	Calculate the Exposure Perimeter (list and score) of target nodes over path or edge data.
graph_node_centrality_fl()	Calculate various metrics of node centrality (such as degree and betweenness) over graph data (edge and nodes).
graph_path_discovery_fl()	Discover valid paths between relevant endpoints (sources and targets) over graph data (edge and nodes).

KQL Graph – what happened since KustoCon 24?

Massive usage & scale growth across models, snapshots, and query scope

300M

Graph Queries

30K+

Graph Models

400K+

Graph Snapshots

4T

Elements Persisted

7.5T

Elements Queried

KQL Graph – Announcement for openCypher

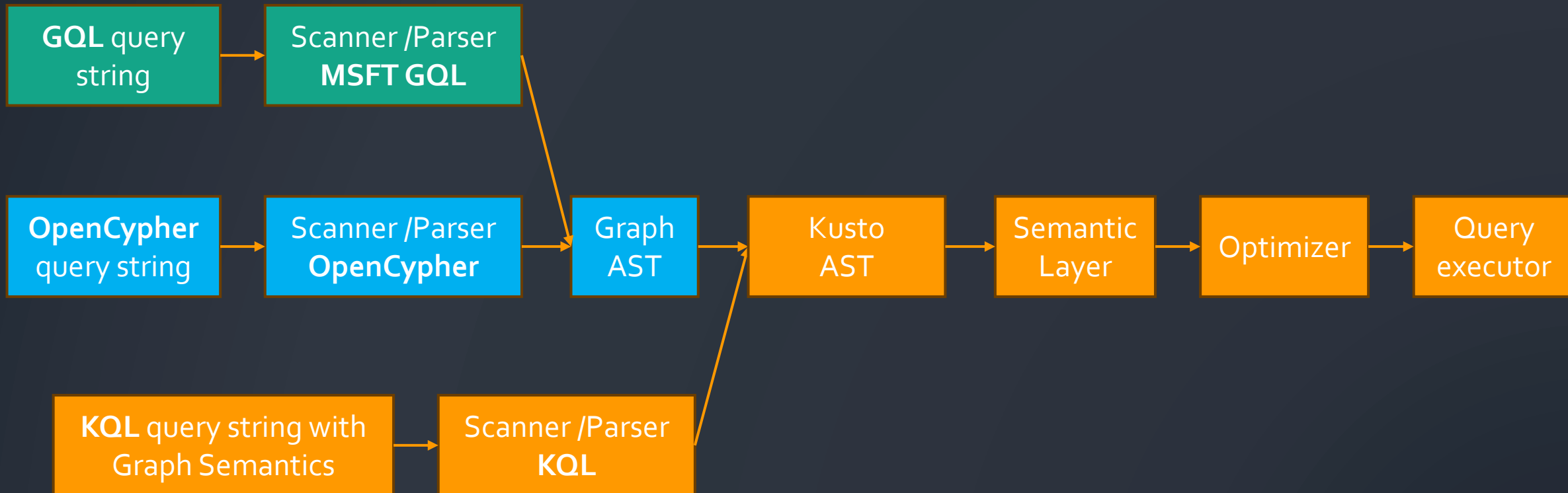
The screenshot displays the Azure Data Explorer interface. On the left, the 'Connections' pane shows a tree view with 'help' and 'Samples' expanded. The 'Samples' folder contains 'Functions', 'Materialized views', and 'Tables'. The 'Tables' folder is selected, showing a list of tables: 'ADX_Conferences', 'Covid19', and 'CyberSecurity'. The main query editor on the right contains a KQL query. The query is as follows:

```
15
16 #crp query_language = opencypher
17
18 #crp query_graph_reference = G_doc()
19
20 #crp query_graph_label_name = lbl
21
22 MATCH (center)-[*1..3]->(connected:Person)
23 WHERE center.name = 'Alice'
24 RETURN DISTINCT connected.name
25 ORDER BY connected.name
```

Below the query editor, the 'Table 1' tab is active, showing the results of the query. The results are displayed in a table with one column, 'connected.name', and five rows of data:

connected.name
Alice
Bob
Carol
David
Emma

KQL Graph – Announcement for openCypher



AI

- MCP – provide tools to access Kusto data plane
 - Fabric RTI MCP (Kusto + all RTI)
 - Azure MCP (Kusto + all Azure)
 - Many other MCP server from the community
- LangGraph checkpoints
 - Mutual learning
 - Scalable, reliable, and restartable foundation for multi-agent scenarios
- NL2KQL

Kusto Terminal

Connections

▼ kuskus

Kuskus

Kuskusppe

KuskusConfidentialppe

KuskusConfidential

Ctrl+N: New

Ctrl+E: Edit

Del: Delete

Space: Refresh

Query Editor

Connected: kuskus | Kuskus

Usage

| count

KustoLogs

| count

DataIngest

| where Timestamp > ago(1d)

| summarize sum(RowCount) / pow(10, 14)

DataIngest

| where Timestamp > ago(5d)

| summarize sum(OriginalSize), sum(CompressedSize) by Source, bin(Timestamp, 1d)

| top 1 by sum_OriginalSize desc

| project ['Max ingestion per cluster on a single day'] = format_bytes(sum_OriginalSize), ['Compressed Size'] = format_byt

DimClustersMv

| where State == "Running"

| extend cores = GetCoresFromSKU(MachineSKU) * MachineCount

| summarize sum(cores)

F5: Execute query | F12: Maximize/Restore

Results

Rows: 1 | Columns: 1 | Duration: 4268ms | ClientRequestId: KustoTerminal;840df12a-9471-4e6f-8810-2a672bd9bbf6

Column1

5.12210571839426

/: Filter | Ctrl+L: Select Columns | Ctrl+R: Row Select | Ctrl+J: JSON Viewer | Ctrl+S: Export | F12: Maximize/Restore

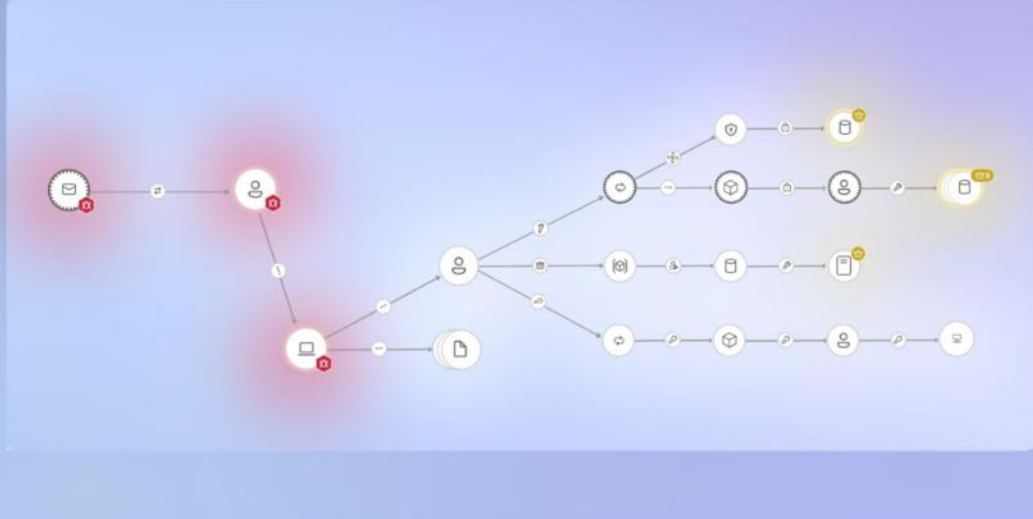
Kusto Terminal | Ctrl+Q: Quit | Alt+Cursor: Switch between frames

<https://github.com/basaba/kusto-terminal>

So what does all of this have to do with the title?

Shawn Bice • 1st
Corporate Vice President, Security Platform & AI, Microsoft | Board Me...
46m •

The term 'graph' is used broadly in the security industry, yet it is often misunderstood or used inaccurately. I thought it might be helpful to share what exactly is a graph, and why is it so important in today's threat landscape.



What Is a Security Graph? Understanding the Foundation of Modern Cybersecurity

Shawn Bice

T-Mobile

Jake Duddy • 2nd
Microsoft MVP - Data Platform | Fabric Super User | Lead Met...
4mo •

In a couple of previous posts, I detailed how to get permission data into a graph structure with Graphframes and visualize it in Power BI using a force-directed Deneb visual. This approach is great for understanding permission inheritance from the perspective of Fabric Items, but it has a significant drawback when you try to view the graph in reverse, from the User's perspective.

This post introduces a better way. By swapping the VertiPaq engine for a Fabric Kusto database, we can leverage its native graph processing capabilities to solve the subgraph filtering problem.

<https://lnkd.in/eR26cpGP>

#PowerBI #MicrosoftFabric #Deneb

Smarter Subgraphs - Kusto as a Engine For Power BI
evaluationcontext.github.io

We are building Kusto and KQL to accelerate **your** innovation

Call to action

- It's never too late to start with <https://detective.kusto.io>
- Learn it all:
 - Eventhouse <https://learn.microsoft.com/training/paths/explore-real-time-analytics-microsoft-fabric/>
 - Azure Data Explorer <https://learn.microsoft.com/en-us/azure/data-explorer/data-explorer-overview>
- Try it
 - Free cluster: <https://dataexplorer.azure.com>
 - Fabric trial: <https://learn.microsoft.com/fabric/get-started/fabric-trial>

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Thank you!



Questions?